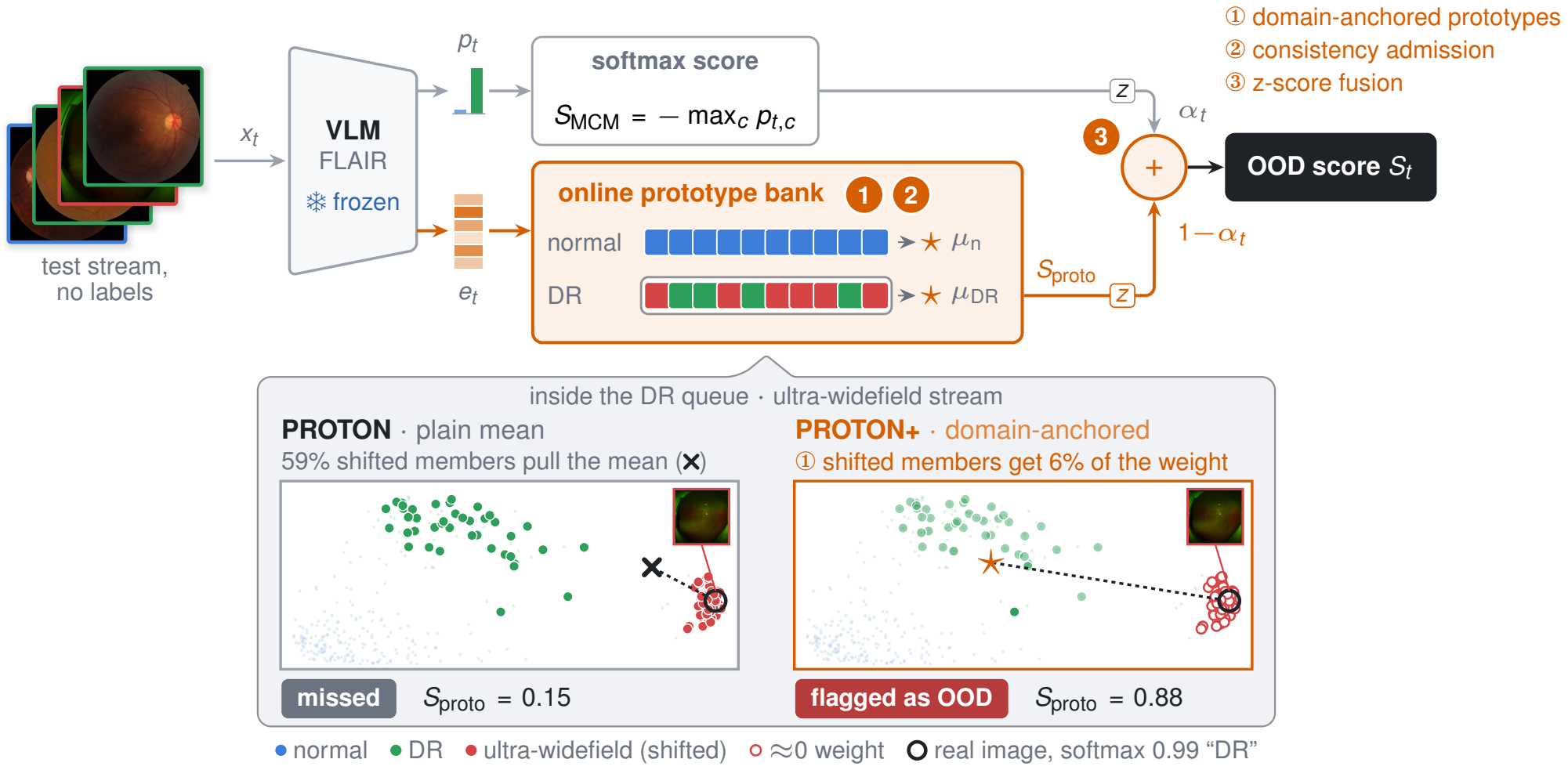

① domain-anchored prototypes
② consistency admission
③ z-score fusion
test stream, no labels
x_t
VLM
FLAIR
❄ frozen
p_t
e_t
softmax score
$S_{MCM} = -\max_c p_{t,c}$
online prototype bank
1
2
normal
DR
μ_n
μ_{DR}
S_{proto}
z
α_t
3
+
OOD score S_t
$1-\alpha_t$
inside the DR queue · ultra-widefield stream
PROTON · plain mean
59% shifted members pull the mean (✕)
missed
$S_{proto} = 0.15$
PROTON+ · domain-anchored
① shifted members get 6% of the weight
flagged as OOD
$S_{proto} = 0.88$
normal
DR
ultra-widefield (shifted)
≈0 weight
real image, softmax 0.99 “DR”